

#This code will be used to demonstrate theorem 6.20. We will demonstrate that mI^2

$$= \frac{-(z+w)^2 - y^2 \cdot ((w+z)^3 - y^3)^2 \cdot ((x^3 - z^3) \cdot w^2 + (x^3 - w^3) \cdot z^2)^2}{z \cdot w \cdot (w+z)^4 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3))^2} \text{ is increasing along}$$

the γ_2 curve; (Part 3 – calculating the sign of the derivative in two points)

restart;

#Define m1

$$m1 := \left(\left(a^2 b^2 + \frac{1}{3} \right)^2 (a+b)^4 (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 + b^3 + \right. \\ \left. + \frac{b(b^4 - 4 b^2 + 1)(b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1)(b^8 + 8 b^4 + 1) a^6}{6} \right. \\ \left. - \frac{b(b-1)^2(b+1)^2(b^2 + 1)^3 a^5}{2} - \frac{b^2(b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \right. \\ \left. - \frac{b^3(b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right)^2 (a^2 + 1)(b^2 + 1)(ba + 1)^2 \Bigg)$$

$$m1 := \left(\left(a^2 b^2 + \frac{1}{3} \right)^2 (a+b)^4 (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 + b^3 + 2 b) a^6 + (3 b^8 + 2 b^6 - 20 b^4 - 4 b^2 + 1) a^5 + (-b^9 + 12 b^5 - 6 b^3 - 3 b) a^4 + (\right. \\ \left. - 2 b^8 - 3 b^6 + 2 b^4 + 3 b^2) a^3 + (-3 b^7 - 5 b^3) a^2 + (-2 b^6 + 3 b^4) a - b^5)^2 \right) / \\ \left(\left((b^8 - b^6 + b^4) a^{10} - \frac{b^2(b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \right. \\ \left. \left. + \frac{b(b^4 - 4 b^2 + 1)(b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1)(b^8 + 8 b^4 + 1) a^6}{6} \right. \right. \\ \left. \left. - \frac{b(b-1)^2(b+1)^2(b^2 + 1)^3 a^5}{2} - \frac{b^2(b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \right. \right. \\ \left. \left. - \frac{b^3(b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right)^2 (a^2 + 1)(b^2 + 1)(ba + 1)^2 \right)$$

#Define r2

$$r2 := \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + (\right. \\ \left. - 4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \right. \\ \left. \left. + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \right) b^8 a^{24} - 4 b^2 (b^{28} \right. \\ \left. + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \right)$$

$$\begin{aligned}
& + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\
& - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\
& - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \right. \\
& + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\
& - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \left. \right) \\
& a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\
& - 27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} \\
& - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\
& - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\
& + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\
& - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\
& - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\
& + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
& - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
& - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\
& + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\
& + \frac{8}{5} b^2 - 1 \left. \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big):
\end{aligned}$$

#Calculating the derivative of $m1$ at $b=1$

$$d := \text{eval}(\text{diff}(m1, a), b=1) \cdot \left(-\frac{\text{eval}(\text{diff}(r2, b), b=1)}{\text{eval}(\text{diff}(r2, a), b=1)} \right) + \text{eval}(\text{diff}(m1, b), b=1) :$$

#Locate a range for $\mathcal{V}(1)$

$$\text{sturm}\left(\text{sturmseq}(\text{eval}(r2, b=1), a), a, \frac{2662}{1000}, \frac{2663}{1000}\right); \text{\#number of roots in } [2.662, 2.663)$$

1

(2)

#Define numerd

$$\text{numer}d := \text{numer}(d)$$

$$\begin{aligned} \text{numer}d := & -(3a^2 + 1)(a + 1)(a^9 - a^8 - 8a^6 - 18a^5 + 2a^4 - 8a^2 + a - 1)a(36a^{54} \\ & + 36a^{53} - 45a^{52} - 204a^{51} - 24a^{50} - 1200a^{49} - 11723a^{48} - 5072a^{47} + 41336a^{46} \\ & + 1129416a^{45} + 590010a^{44} - 1520312a^{43} - 20217584a^{42} + 3267856a^{41} - 20094690a^{40} \\ & + 69858864a^{39} - 158090836a^{38} + 339484700a^{37} + 15879169a^{36} + 151831084a^{35} \\ & + 110537784a^{34} - 1309974880a^{33} + 1180193679a^{32} - 2246830240a^{31} + 2780960464a^{30} \\ & - 2212903184a^{29} + 2211608380a^{28} - 1198680976a^{27} + 304479584a^{26} + 853142688a^{25} \\ & - 1659280012a^{24} + 1989876448a^{23} - 2234366788a^{22} + 2024599196a^{21} - 1609642843a^{20} \\ & + 1127481548a^{19} - 708510376a^{18} + 381869584a^{17} - 184357117a^{16} + 40902256a^{15} \\ & - 5688520a^{14} - 13106744a^{13} - 2137494a^{12} + 228040a^{11} - 2050160a^{10} - 401840a^9 \\ & + 152718a^8 - 250640a^7 + 13812a^6 + 3236a^5 - 8921a^4 + 340a^3 + 136a^2 + 25) \end{aligned} \quad (3)$$

#Define denomd

$$\text{denom}d := \text{denom}(d)$$

$$\begin{aligned} \text{denom}d := & 2(3a^{10} - 9a^8 - 24a^7 + 10a^6 - 6a^4 - 8a^3 + 3a^2 - 1)^3(a^2 + 1)^2a^3(a^{28} - 7a^{24} \\ & - 208a^{22} + 405a^{20} + 23552a^{19} + 1232a^{18} + 10752a^{17} - 319075a^{16} - 48640a^{15} \\ & - 592416a^{14} - 26112a^{13} - 570845a^{12} + 23040a^{11} - 398944a^{10} + 19968a^9 - 143253a^8 \\ & + 5632a^7 - 42000a^6 - 4608a^5 + 647a^4 - 3584a^3 + 144a^2 - 65) \end{aligned} \quad (4)$$

#Verify that numerd and denomd have the same sign in the interval.

$$\text{sturm}\left(\text{sturmseq}(\text{numer}d, a), a, \frac{2662}{1000}, \frac{2663}{1000}\right); \text{\#number of roots in } [2.662, 2.663);$$

$$\text{sturm}\left(\text{sturmseq}(\text{denom}d, a), a, \frac{2662}{1000}, \frac{2663}{1000}\right); \text{\#number of roots in } [2.662, 2.663)$$

0

0

(5)

#Calculate the sign of $mI'(\gamma_2(1), 1)$ using $mI'\left(\frac{2662}{1000}, 1\right)$

$$\frac{\text{eval}\left(\text{numerd}, a = \frac{2662}{1000}\right)}{\text{eval}\left(\text{denomd}, a = \frac{2662}{1000}\right)}$$

91466949515065458179816951179460033959629438066603109972258793046721203978030880\ (6)
 93658953089816296403870440432405862911752192744533072787775763393029637296475\
 396845090812760899204858628551392429875003119/
 45468649053779655688622403244264004471306988107315889544897932958953154365757\
 23064921392934242052288300938703395017962414557365880252023919606145408384794\
 5676052478029644474506369880096224584390125000

restart;

$$\begin{aligned}
 m1 := & \left(\left(a^2 b^2 + \frac{1}{3} \right)^2 (a+b)^4 (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 \right. \\
 & + b^3 + 2 b) a^6 + (3 b^8 + 2 b^6 - 20 b^4 - 4 b^2 + 1) a^5 + (-b^9 + 12 b^5 - 6 b^3 - 3 b) a^4 + (\\
 & \left. -2 b^8 - 3 b^6 + 2 b^4 + 3 b^2) a^3 + (-3 b^7 - 5 b^3) a^2 + (-2 b^6 + 3 b^4) a - b^5)^2 \right) / \\
 & \left(\left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \right. \\
 & + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\
 & - \frac{b (b-1)^2 (b+1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\
 & \left. \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right)^2 (a^2 + 1) (b^2 + 1) (b a + 1)^2 \right) \\
 m1 := & \left(\left(a^2 b^2 + \frac{1}{3} \right)^2 (a+b)^4 (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 \right. \\
 & + b^3 + 2 b) a^6 + (3 b^8 + 2 b^6 - 20 b^4 - 4 b^2 + 1) a^5 + (-b^9 + 12 b^5 - 6 b^3 - 3 b) a^4 + (\\
 & \left. -2 b^8 - 3 b^6 + 2 b^4 + 3 b^2) a^3 + (-3 b^7 - 5 b^3) a^2 + (-2 b^6 + 3 b^4) a - b^5)^2 \right) / \\
 & \left(\left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \right. \\
 & + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\
 & - \frac{b (b-1)^2 (b+1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\
 & \left. \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right)^2 (a^2 + 1) (b^2 + 1) (b a + 1)^2 \right)
 \end{aligned}$$

#Define r2

$$\begin{aligned}
 r2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + (\right. \\
 & -4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\
 & + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \left. \right) b^8 a^{24} - 4 b^2 (b^{28} \\
 & + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12}
 \end{aligned}$$

$$\begin{aligned}
& + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\
& - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\
& - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \right. \\
& + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\
& - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \left. \right) \\
& a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\
& - 27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} \\
& - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\
& - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\
& + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\
& - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\
& - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\
& + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
& - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
& - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\
& + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\
& + \frac{8}{5} b^2 - 1 \left. \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big):
\end{aligned}$$

#Calculating the derivative of $m1$ at $b=1.4$

$$d := eval\left(diff(m1, a), b = \frac{7}{5}\right) \cdot \left(- \frac{eval\left(diff(r2, b), b = \frac{7}{5}\right)}{eval\left(diff(r2, a), b = \frac{7}{5}\right)} \right) + eval\left(diff(m1, b), b = \frac{7}{5}\right) :$$

$$\begin{aligned} & \# \text{Locate a range for } \gamma_2 \left(\frac{7}{5} \right) \\ & \text{sturm} \left(\text{sturmseq} \left(\text{eval} \left(r_2, b = \frac{7}{5} \right), a \right), a, \frac{21}{10}, \frac{22}{10} \right); \# \text{number of roots in } [2.1, 2.2) \\ & \quad \quad \quad 1 \end{aligned} \tag{8}$$

$$\begin{aligned} & \# \text{Define numerd} \\ & \text{numer}d := \text{numer}(d) : \\ & \# \text{Define denomd} \\ & \text{denom}d := \text{denom}(d) : \\ & \# \text{Verify that numerd and denomd have the same sign in the interval.} \\ & \text{sturm} \left(\text{sturmseq}(\text{numer}d, a), a, \frac{21}{10}, \frac{22}{10} \right); \# \text{number of roots in } [2.1, 2.2]; \\ & \text{sturm} \left(\text{sturmseq}(\text{denom}d, a), a, \frac{21}{10}, \frac{22}{10} \right); \# \text{number of roots in } [2.1, 2.2] \\ & \quad \quad \quad 0 \\ & \quad \quad \quad 0 \end{aligned} \tag{9}$$

$$\begin{aligned} & \# \text{Calculate the sign of } mI' \left(\gamma_2 \left(\frac{7}{5} \right), \frac{7}{5} \right) \text{ using } mI' \left(\frac{215}{100}, \frac{7}{5} \right) \\ & \frac{\text{eval} \left(\text{numer}d, a = \frac{215}{100} \right)}{\text{eval} \left(\text{denom}d, a = \frac{215}{100} \right)} \\ & 9092823565625256694539570500679626286493543369362886706683620205115201639394897 \backslash \tag{10} \\ & 27687162102571539796538975484375179445459749877635916589636397951562500 / \\ & 1536306614875233573204789998955053679534511339388131844049518040768802075532 \backslash \\ & 92238741350075658219750029461034674770912845096020914142168717364391949 \end{aligned}$$